IMPROPER INTEGRALS

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Name: Solutions

Determine whether the following improper integrals converge or diverge.

TYPE I

1. $\int_1^\infty \frac{\ln(x)}{x}\,\mathrm{d}x$

$u = \ln(x)$

$du = \frac{1}{x}dx$

$$= \lim_{t\to\infty} \int_1^t \frac{\ln(x)}{x}\,dx$$

$$= \lim_{t\to\infty} \int_0^{\ln(t)} u\,du = \lim_{t\to\infty} \frac{1}{2}u^2 \Big|_0^{\ln(t)}$$

$$= \lim_{t\to\infty} \frac{1}{2}\left(\ln(t)^2\right) = \boxed{\infty}$$

2. $\int_1^\infty \frac{1}{(2x+1)^3}\,\mathrm{d}x$

$u = 2x+1$

$du = 2dx \Rightarrow \frac{1}{2}du = dx$

$u(1) = 2(1)+1 = 3$

$u(t) = 2t+1$

$$\lim_{t\to\infty} \int_1^t \frac{1}{(2x+1)^3}\,dx = \lim_{t\to\infty} \frac{1}{2}\int_3^{2t+1} u^{-3}\,du$$

$$= \lim_{t\to\infty} \frac{1}{2}\,\frac{u^{-2}}{-2}\Big|_3^{2t+1} = \lim_{t\to\infty} -\frac{1}{4}\left(\frac{1}{(2t+1)^2} - \frac{1}{3^2}\right)$$

$$= -\frac{1}{4}\left(0 - \frac{1}{9}\right) = \boxed{\frac{1}{36}}$$

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3. $\int_{-\infty}^{0} e^x \, dx = \lim_{t \to -\infty} \int_t^0 e^x dx$

$$= \lim_{t \to -\infty} e^x \Big|_t^0$$

$$= \lim_{t \to -\infty} (e^0 - e^t)$$

$$= 1 - 0 = \boxed{1}$$

4. $\int_{-\infty}^{\infty} x e^{-x^2} \, dx = \int_{-\infty}^{0} x e^{-x^2} dx + \int_0^{\infty} x e^{-x^2} dx = -\frac{1}{2} + \frac{1}{2} = \boxed{0}$

$u = -x^2$

$-\frac{1}{2} du = x dx$

$$\int_{-\infty}^{0} x e^{-x^2} dx = \lim_{t \to -\infty} \int_t^0 x e^{-x^2} dx$$

$$= \lim_{t \to -\infty} -\frac{1}{2} \int_{-t^2}^{0} e^u du = \lim_{t \to -\infty} -\frac{1}{2} e^u \Big|_{-t^2}^{0}$$

$$= \lim_{t \to -\infty} -\frac{1}{2}(1 - e^{-t^2}) = -\frac{1}{2}(1 - 0) = -\frac{1}{2}$$

$$\int_0^{\infty} x e^{-x^2} dx = \lim_{t \to \infty} \int_0^t x e^{-x^2} dx = \lim_{t \to \infty} -\frac{1}{2} \int_0^{-t^2} e^u du$$

$$= \lim_{t \to \infty} -\frac{1}{2}(e^{-t^2} - e^0) = -\frac{1}{2}(0 - 1) = \frac{1}{2}$$

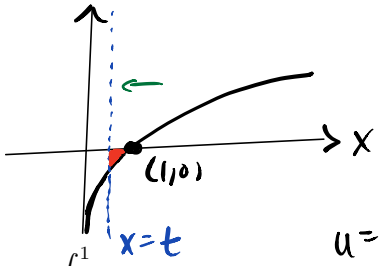
<u>Warning</u>: $f(x) = xe^{-x^2}$ is odd:

$$f(-x) = (-x)e^{-(-x)^2} = -xe^{-x^2} = -f(x).$$

$\int_{-a}^{a} (\text{odd function})\,dx = 0$, but this <u>not</u> true for improper integrals!

$$\int_{-\infty}^{\infty} (\text{odd function})\,dx \neq 0$$

E.g.: Both $\int_0^{\infty} x\,dx$ and $\int_{-\infty}^{0} x\,dx$ diverge, so $\int_{-\infty}^{\infty} x\,dx$ diverges (not $=0$).



TYPE II

5. $\int_0^1 \ln(x)\,\mathrm{d}x$

$u = \ln(x) \quad v = x$

$du = \frac{1}{x}dx \quad dv = dx$

$$\lim_{t\to 0^+} \int_t^1 \ln(x)\,dx = \lim_{t\to 0^+}\left(x\ln(x)\Big|_t^1 - \int_t^1 dx\right)$$

$x\Big|_t^1$

$$= \lim_{t\to 0^+}\left(0 - t\ln(t) - (1-t)\right)$$

$$= \lim_{t\to 0^+}\left(-t\ln(t) - 1 + t\right)$$

$$= \boxed{-1}$$

$$\lim_{t\to 0^+} \frac{\ln(t)}{\frac{1}{t}}$$

$$\text{L'H} = \lim_{t\to 0^+} \frac{\frac{1}{t}}{-\frac{1}{t^2}}$$

$$= \lim_{t\to 0^+} \frac{-t^2}{t}$$

$$= \lim_{t\to 0^+} -t = 0$$

6. $\int_0^1 \frac{\mathrm{d}x}{\sqrt{x}}$

$$\int_0^1 \frac{dx}{\sqrt{x}} = \lim_{t\to 0^+} \int_t^1 x^{-1/2}dx = \lim_{x\to 0^+} 2\sqrt{x}\Big|_t^1$$

$$= \lim_{x\to 0^+} 2\left(\sqrt{1} - \sqrt{t}\right)$$

$$= 2(1-0)$$

$$= \boxed{2}$$